

MIRROR PAGE

A comic-page light puzzle where the narrator breaks the rules. TGC Game Jam proposal.

Team	Sankeerth Nara (solo)
Repository	https://github.com/SankeerthNara/TGC-Game-Jam (public, MIT license)
Themes	Comic, Twist, Light (all three, in one game)
Engine / platform	Godot 4.7, GDScript, Compatibility renderer; browser (HTML5) build on itch.io
Target playtime	15 minutes first playthrough (20 minutes maximum)

1. Pitch

Every level is a page of a comic book. A beam of light starts in one panel and must reach the stars on the page. You solve it by **dragging whole comic panels to swap them** and **flipping mirrors**. Because emitters, walls and targets live inside panels, rearranging the page rearranges the puzzle. An unreliable narrator speaks in caption boxes, and keeps breaking the rules of the world.

2. How the three themes work together

Theme	How it appears in the game
Comic	The page is the puzzle. Panels are the movable pieces, gutters are rule-changing borders, narrator captions tell a story, panels shake with comic impact effects, and sound effects appear as comic words (POW, ZAP).
Light	The beam of light is the only mechanic. It reflects off mirrors, stops at walls and lights targets. It is the player's only tool and the only feedback.
Twist	The narrator twists the game itself. (1) Goal twist : after you solve a page, the caption reveals the hero you were lighting is the villain. Good and bad targets swap and you must re-solve the same page. (2) Rule twists : mirrors start lying, and the gutters start bending light.

3. Core mechanics

- **Swap panels**: drag one panel onto another. Everything inside moves with it.
- **Flip mirrors**: click a mirror to switch its angle (/ or \).
- **Beam**: reflects off mirrors, is blocked by walls, ends on a target. A level is solved when every good target is lit and no bad target is lit.
- **Locked panels**: some panels cannot be moved, which constrains the solution in later levels.
- **Undo / reset**: unlimited undo, so players can experiment freely.

4. Rule twists (what makes it unique)

Rule	Effect	Level
Normal	Mirrors reflect as drawn.	1, 2, 3
Goal flip	After the first solve, stars become villains and red X targets become the goal.	3 and 6
Lying mirrors	Every mirror reflects the opposite way to what it shows.	4

Gutter bend	A beam entering a different panel turns 90 degrees clockwise. Panel order now decides the path.	5, 6
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5. Gameplay loop and content (about 15 minutes)

#	Page	Teaches / new idea	Time
1	Page One	Swapping panels to carry light	1.5 min
2	Looking Glass	Mirrors combined with panel swaps	2.5 min
3	The Twist	Goal twist: hero becomes villain	2.5 min
4	Liar, Liar	Lying mirrors	3 min
5	Gutter Bend	Beam bends at panel borders, locked panels	3 min
6	The Last Page	All rules together plus final flip, then the ending	3 min

Loop per page: read the caption, rearrange the page, light the stars, watch the narrator react, turn the page. Pages get a short intro caption, a solve caption, and a twist caption where relevant.

6. Final scope

In scope: panel swapping, mirror flipping, undo and reset, locked panels, three rule types, the goal-flip twist, narrator captions, 6 levels, main menu, pause menu, ending screen, music and sound effects, comic-style art, browser build on itch.io.

Out of scope (cut first if time runs short): level editor, save system, more than 6 levels, controller and mobile touch support, localisation.

Success criteria: a new player finishes all 6 levels in 15 to 20 minutes with no instructions beyond the in-game captions, and the web build runs without errors in a browser.

7. Technical plan

- **Architecture:** a pure-logic model (PageModel, BeamSolver) separated from the view (PageView) and game flow. UI, audio and effects only listen to a global signal bus. This keeps the logic testable without running the game.
- **Levels as data:** JSON files describing panels and cells. A built-in headless **level validator** brute-forces every panel arrangement to prove each level is solvable and reports its minimum number of moves; a level generator proposes candidate layouts.
- **Web safety:** Compatibility renderer and GDScript only (required for Godot web export); no threads or native plugins. Web builds are tested in a browser regularly, not only at the end.
- **Status at proposal time:** a playable core prototype already exists in the repository (panel swapping, mirrors, beam, goal-flip twist, 3 levels, validator), with placeholder visuals.

8. Milestones (100 hours)

Hours	Goal
0 - 12	Concept and scope lock; core prototype; documentation; this proposal.
12 - 30	Vertical slice: menu, HUD, caption boxes, first art and audio pass, levels 1-3 polished, web export working.

30 - 60	Levels 4-6, lying and bend rules, locked panels, full comic art and sound, music.
60 - 85	Polish, game feel, difficulty tuning from playtests, tutorial captions, bug fixing.
85 - 100	Feature freeze at hour 85, final web build, itch.io page, README and CREDITS, submission before the deadline.

9. Assets and AI disclosure

- **Assets:** only free, open-source, CC0 / CC BY or AI-generated assets. No paid assets. Every third-party asset is listed in CREDITS.md with source and license, and on the itch.io page.
- **AI tools used:** Claude Code (architecture, core gameplay code, review), Antigravity (UI, art and audio generation, level authoring, browser testing) and ChatGPT / Codex (tests, documentation, small isolated features). All use is logged in docs/AI_USAGE.md.
- **Human direction:** concept selection, design decisions, level and narrator writing direction and all merges are done by the developer; reasoning is recorded in the design log. Workflow and ownership between the tools is documented in AI_WORKFLOW.md.
- **Editor tooling:** the open-source Godot AI plugin (MIT) is used in the editor only and is not part of the game.

10. Risks and mitigations

Risk	Mitigation
Levels too easy, too hard or unsolvable	Automatic solvability and minimum-move check on every level; playtests; undo and hints.
Web build problems	Compatibility renderer from day one; browser test of every milestone; small asset budget.
AI quota limits	Work split across three tools with owned folders and a shared task board with handoff notes.
Scope creep	Fixed 6-level scope; cut list defined above; feature freeze at hour 85.